

Relationship between fractal feature and compressive strength of concrete based on MIP

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ABSTRACT

Concrete is a heterogeneous, multiphase, and multilayer composite system with extremely intricate microscopic pore structure. The structural characteristics of microscopic pore of concrete are closely related to its mechanical properties and durability. The pore structure of concrete with various strength grades (C10-C150) was analyzed by Mercury intrusion porosimetry (MIP) from the aspects of pore diameter distribution, pore volume, pore specific surface area and fractal feature. The Menger sponge model was used to calculate the fractal dimension of different pore regions in each test group, and the most suitable pore fractal region was selected to establish the mathematical model between fractal dimension and compressive strength. The results showed that all fractal curves are divided into four segments according multi-fractal characteristic of concrete, namely Region I, Region II, Region III and Region IV. Region I is macropores fractal region ($d = 10 \mu\text{m} \sim 950 \mu\text{m}$), representing the packing patterns of hydrated cement grains, while Region II - Region IV is micropores fractal region ($d = 0.0064 \mu\text{m} \sim 10 \mu\text{m}$), reflecting mainly the microstructure of C-S-H hydrates. Region I and Region IV exhibit obvious fractal features and the fractal dimension D is between 2 and 3. However, the fractal dimension $D > 3$ in Region II and Region III, which are nonphysical from the point of view of surface geometry. The fractal dimension can clearly describe the internal pore characteristic, and it is a real good correlation with the most probable pore diameter and the pore specific surface area. Moreover, the mathematical model between the fractal dimension and the compressive strength both accords with the positive power exponential function in Region I and Region IV. The fractal dimension can be viewed as a comprehensive parameter of pore shape and spatial distribution of pore structure, which can more accurately represent the relationship between the microscopic pore structure features and compressive strength of concrete. In practical engineering, it serves to realize the damage detection of concrete on site.

1. Introduction

Concrete has become the most widely used building material today due to its superior performance and wide range of materials [1]. Concrete is a heterogeneous, multiphase, and multilayer composite system with extremely complex microscopic pore structure [2]. The macroscopic characteristics of concrete are the reflection of its complexity microstructural, such as irregularity, indeterminacy, and nonlinearity. And the structural characteristics of macroscopic agglomerations and microscopic pore of concrete are closely related to its mechanical properties and durability. For instance, mechanical properties, density, thermal conductivity, volume stability, hygroscopicity and permeability of materials are directly or indirectly related to its porosity. Moreover,

the shape, diameter size, surface area and spatial distribution of pores are also influencing factors [3-7]. However, there are different pores diameter in concrete, such as residual bubbles, capillary pores, gel pore and fine cracks caused by drying shrinkage, which results in random disorder and diversity of the morphology and spatial distribution of pores, and makes the micro-structure of cement composites become abnormal chaotic and complex which traditional Euclidean geometry cannot accurately describe. According to fractal theory, the pores of porous materials have obvious fractal features in its pore volume, pore surface area and pore size distribution [8-13]. Fractal dimension is a parameter that can be used to mark the complexity of a fractal set, and each fractal set corresponds to a fractal dimension defined in some way. Fractal dimension has broken through the boundaries of integer

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dimension in general topological set, which can be expressed by fraction and can describe the essential characteristics of objects. This is also the basic point that distinguishes it from classical geometry. The fractal dimension can be used to describe the complexity of the morphology and spatial distribution of pores when the fractal theory is applied to represent the microscopic pore structure of concrete. The larger the fractal dimension, the more complex of the morphology and spatial distribution of pores in concrete. Studies have shown that [14,15], the fractal dimension can play a bridge role between the microstructure and the macroscopic performance of the concrete, and it can be a quantitative indicator for guiding material design. Therefore, it is very effective to use fractal theory to analyze and evaluate the mechanism of new concrete, to describe the self-similarity of microscopic structure, mechanical properties under the microscopic level and the macroscopic field [16,17].

Previous studies have shown that the concrete contains a large number of pores and microcracks of different orders, which are very similar to the sponge [18,19]. In addition, the construction principle of the Menger sponge is similar to the measuring principle of MIP. Therefore, the fractal model of Menger sponge can be utilized to simulate the pore characteristics between material particles. Many researchers have studied the pore structure of cement-based compound cementitious materials and established a fractal calculation model based on the construction principle of the Menger sponge model [9,18–23]. Friesen [22] constructed the surface fractal model of porous materials based on the test results of MIP and the construction principle of Menger sponge, and calculated the pore surface fractal dimension of porous materials. Ji et al. [23] studied the relationship between pore volume and pore size, and established a calculation model for the fractal dimension of pore volume. And Ji considered that the construction process of the Menger sponge is analogous to the process of filling pores with the hydration product of compound cementitious materials.

At present, although some researchers have studied the range of fractal dimension and the relationship between fractal dimension and mechanical properties of concrete [19,24–27]. But the conclusions reached are quite different and some are even opposite [20,25,26]. According to the current research, it is found that there is still not accurate conclusion on the reasonable range of fractal dimension of microscopic pore structure and the relationship between fractal dimension and strength in concrete. In addition, there is little systematic expression in the mathematical model that represents the relationship between microscopic pore structure parameters and compressive strength beyond the tensile or the combined strength of a particular material. Therefore, this paper constructs a reasonable fractal model based on the microscopic pore structure characteristics of concrete. The pore structure of 11 kinds of concrete with different strength grades was analyzed by mercury intrusion porosimetry (MIP) on the aspects of pore diameter distribution, pore volume, specific surface area and fractal feature. According to the multifractal nature of the fractal curve, the Menger sponge model was used to calculate the fractal dimension of different pore stages in each test group. Then, multiple characterization parameters of pore structure were considered include the pore volume, pore diameter distribution and specific surface area, the optimal parameters were selected to establish the mathematical model between microscopic pore structure and compressive strength.

2. Materials and experiments

2.1. Raw materials

The raw materials used in concrete include P-O 42.5 Portland cement (C) and silica fume (SF), aggregates and water. The chemical compositions of cement and silica fume are exhibited in Table 1. And the physical properties of cement and silica fume are listed in Table 2 and Table 3. Aggregates consist of crushed limestone with particle size of 5–20 mm and river sand with fineness modulus of 2.8. Clean tap water and polycarboxylate superplasticizer with 19.1% water reducing rate are applied in mixing.

2.2. Mixture proportions and preparation of specimens

For concrete, the water-binder ratio (W/C) is one of the most important factors affecting its compressive strength. Table 4 shows the mix proportion of 11 different strength grades concrete, which based on the Chinese national standard “Specification for mix proportion design of ordinary concrete (JGJ 55–2011)” [28].

Table 5 shows the forming and curing methods of specimens, according to a Chinese Standard “Standard for test method of mechanical properties on ordinary concrete (GBT50081–2002)” [29].

2.3. Mercury intrusion porosimetry (MIP)

The Mercury intrusion porosimetry (MIP) of concrete specimens after curing 7, 28, 90 days were conducted by PoreMaster 33 mercury porosimeter in the pressure range of 0.23 ~ 33000 Psia, which measured pores diameters was 0.0064 ~ 950 μm . Besides, the contact angle between the mercury and the pore surface was $\theta = 130^\circ$. The size of MIP fragment specimens was 3–5 mm approximately, taken from the concrete specimen cores. Before tested, MIP specimens were placed in absolute ethanol for at least 24 h to dehydrate and terminate hydration. Then, the specimens were put into the oven under 60°C until it achieved constant mass to evaporate the free water and alcohol.

2.4. Compressive strength experiment

The compressive strength experiment was carried out at curing ages of 3 days, 7 days, 28 days, 90 days according to a Chinese Standard “Standard for test method of mechanical properties on ordinary concrete (GBT50081–2002)” [29]. Triplicate specimens were tested for each concrete mixture, and their average value were taken as the final value of the strength.

3. Fractal model

3.1. The establishment of fractal model

The concrete contains a large number of pores and microcracks of different orders, which are very similar to the sponge [18]. The fractal model of Menger sponge can be utilized to simulate the pore characteristics between material particles [22]. The fractal model of concrete was constructed based on the fractal theory of Menger sponge [27,30]. Fig. 1 shows the geometric constructed process of Menger Sponge.

The construction process is as follows: Firstly, the cube is divided with side length R into m^3 small cubes. According to a selected iteration rule, n sub-cubes are randomly removed and N_1 ($N_1 = m^3 - n$) sub-cubes

Table 1
Chemical compositions of binder (%).

Raw materials	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	SO ₃	Na ₂ O _{eq}	Cl ⁻	Loss
P.O 42.5 cement	20.85	4.93	3.40	61.85	2.69	2.53	0.78	0.05	3.45
Silica fume	94.86	1.69	0.35	0.26	0.20	—	—	—	1.00

Table 2
Physical properties of cement.

Materials	Fineness		Normal consistency(%)	Initial setting time(min)	Final setting time(min)	Compressive strength(MPa)		Flexural Strength(MPa)		Stability
	Specific surface area(m ² /kg)	Residue on 45 μm square hole sieve(%)				3 days	28 days	3 days	28 days	
P.O 42.5 Cement	353	6.3	27.9	218	275	26.4	54.2	5.3	8.2	8.2

Table 3
Physical properties of silica fume.

Materials	Water content (%)	Ignition loss(%)	Density (g/cm ³)	Specific surface area (m ² /kg)	28d active index (%)
Silica fume	0.06	5.48	2.1	13,000	131

remain. The remaining sub-cubes are then repeatedly iterated according to the above process. Each of the remaining sub-cubes together constitutes the matrix of the material, and the removed sub-cubes with different sizes are equivalent to pores or microcracks with different diameter inside the material. After k times of iteration, the radius r_k and number N_k of remained sub-cubes is expressed respectively following Eq (1) and (2).

$$r_k = R/m_k \quad (1)$$

$$N_k = N_1^k = (m^3 - n) = (R/r_k)^{lgN_1/lgm} \quad (2)$$

According to Hausdorff dimension, the pore volume fractal dimension D of Menger sponge is expressed as.

$$D = lgN_1/lgm \quad (3)$$

The volume of remained sub-cubes can be expressed as.

$$V_k = r_k \cdot N_1^k = r_k^{3-D} \quad (4)$$

When $k \rightarrow \infty$, $r_k \rightarrow 0$, the cumulative pore volume of V_p is expressed as follows.

$$V_p = R^3 - V_k \quad (5)$$

The fractal model of Menger sponge can be expressed as.

$$\log[-dV_p/dr] \sim (2-D)\log r \quad (6)$$

The pore volume of concrete can be measured using the MIP by changing the pressure exerted on liquid mercury [12,30]. To overcome the surface tension between mercury and solids, the pressure P must be applied to the radius is r , before the mercury is injected the pores. The relationship between P and r in cylindrical pores satisfies the Washburn:

$$P = \frac{2T\cos\theta}{r} \quad (7)$$

Table 4
Mix proportion of concrete specimens.

Samples	W/C	Sand ratio(%)	Water(kg/m ³)	Cement(kg/m ³)	Silica fume(kg/m ³)	Sands(kg/m ³)	Stones(kg/m ³)	Water reducing agent(kg/m ³)
C10	0.52	38	104	200	—	800	1300	—
C20	0.48	38	96	200	—	800	1300	0.116
C30	0.44	38	176	400	—	700	1125	—
C40	0.38	38	152	400	—	700	1125	0.116
C50	0.34	38	136	400	—	700	1125	0.166
C60	0.3	38	120	400	—	700	1125	0.166
C70	0.28	38	186.76	600	67	550	800	0.266
C80	0.26	38	173.42	600	67	550	800	0.266
C100	0.22	38	146.74	600	67	550	800	0.316
C150	0.18	38	120.06	600	67	550	800	0.316

Where T is the surface tension of mercury, $T = 0.458 \text{ N/m}$; θ is the contact angle between mercury and solid, $\theta = 130^\circ$; P is the pressure applied when mercury is injected into the pores.

At a certain pressure, the total pore volume is equal to the volume of mercury is injected into the pores. Combining Eq. (6) and (8), the fractal dimension of concrete can be calculated determined as follows:

$$\log[-dV_p/dP] \sim (D-4)\log(P) \quad (8)$$

The fractal model of concrete was based on the Menger sponge, which topological dimension is 2. According to the definition of fractal dimension, fractal dimension is strictly larger than its topological dimension. Moreover, in Eq (5), when $V_p = 0$, the fractal dimension $D = 3-3\ln R/\ln r_k = 3$. $D = 2$ and $D = 3$ correspond to smooth surface and volume-filling of concrete [13], respectively. It is nonphysical when $D > 3$ or $D < 2$, from a surface geometry point of view, and the more it approaches 3, the more complex the pore surface is. Therefore, in the fractal model of concrete based on the fractal theory of Menger sponge, the fractal dimension must be in the range $2 < D_3 < 3$ [31].

3.2. The calculation of fractal dimension

The pore structure parameters of concrete can be measured using the MIP by changing the pressure exerted on liquid mercury, such as pore volume V , pore diameter P and specific surface area S , etc. Import the data measured in MIP from poremaster into Excel. First, solve the $\lg(-dV_i/dP_i)$ and $\lg P_i$ by Excel, and draw a scatter diagram with $\lg(-dV_i/dP_i)$ and $\lg P_i$ as Y and X axes, respectively. Then, Adding the linear relation expression to the scatter diagram, the correlation coefficient of R^2 and slope of the linear are obtained by the linear relation expression. Finally, fractal dimension D was calculated by Eq. (8).

Table 5
Forming and curing methods of specimens.

Experiments	Specimen size	Number of samples (block)	Curing condition	Ages
Compressive strength test	100 mm × 100 mm × 100 mm	30	20 ± 2°C RH. ≥ 95%	7 days, 28 days, 90 days
MIP		10		

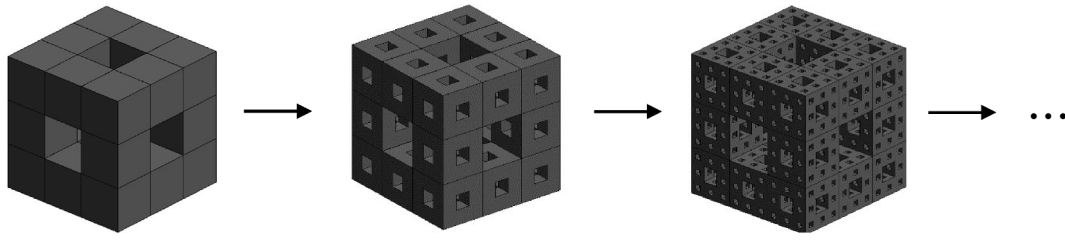


Fig. 1. Menger Sponge.

4. Results and discussion

4.1. Pore diameter distribution

Fig. 2 and Fig. 3 shows that the incremental and cumulative pore diameter distribution curves of each test group at 7 days, 28 days and 90 days. The most probable pore diameter and the threshold pore diameter are used to characterize the changes in the microscopic internal pore structure of concrete in Fig. 2 and Fig. 3, respectively. The most probable pore diameter which is the pore diameter of the largest number of pores in the concrete, while the threshold pore diameter is the pore diameter when the pore volume in concrete increases significantly. The curves move to the left with the decrease of W/C and the pore diameter distribution range of C10-C150 concrete varies significantly. At 7 days, the most probable diameter of C10 concrete is close to that of C20 concrete, which is 1.6, 2.5 and 3.8 times of C60, C80, and C150 concrete respectively. Meanwhile, the threshold pore diameter of C10 and C20 concrete is between 300 nm and 400 nm, while C60-C80 concrete is between 10 nm and 200 nm and C100-150 concrete is between 50 nm and 100 nm. It is caused by excess water forming many pores inside the concrete when the W/C is too large. At 28 days and 90 days, the change trend of pore diameter distribution between C10-C150 concrete is the same as that of 7 days. Moreover, the peak position moves to the left and the peak value of each curve decreases gradually over time, in Fig. 2. This phenomenon means that the most probable diameter and the number of pores in the concrete reduces. Meanwhile, the cumulative pore volume and the threshold pore diameter decrease with curing ages, according to Fig. 3. It indicates that the pore structure refines gradually in the concrete with the progress of hydration. With the curing age, water and cement in concrete react to generate a large number of hydration products, such as C-S-H gel and Ca(OH)₂, which are wrapped on the surface of unhydrated cement particles to form a dense internal microstructure. At the same time, the unhydrated cement particles also play a role in filling the internal pores of concrete. In the later stage of hydration, with the increase of the density of pore structure, the growth space of cement particles is lack, and unhydrated the cement particles hydrated in situ to form very dense hydration products. Therefore, the coarse pores in concrete, microcracks at the interface between aggregate

and cement paste gradually disappear, and denser pore structure is formed inside the concrete. Note that the difference in pore diameter distribution between C10-C150 concrete decreases, at 28 days and 90 days. In addition, this may be due to the hydration process of C10-C150 concrete gradually tends to be stable after 28 days.

4.2. Pore volume distribution

The pore volume distribution of concrete is shown in Fig. 4. The effect of porosity at each pore grade on strength is different. According to the porosity at each pore grade in concrete and their influence coefficient, the pore diameter is finely divided into four categories, as follows: harmless pores ($d < 20$ nm), little harmful pores ($d = 20$ nm–50 nm), harmful pores ($d = 50$ nm–200 nm), and much harmful pores ($d > 200$ nm) by Wu [32]. At 28 days, the total pore volume of C30 and C60 concrete is 1.72 times and 1.37 times that of C150 concrete, respectively. The volume of much harmful pores and harmful holes in C30 concrete accounted for 25.77% and 14.43 % of its total volume, respectively, while accounted for 12.28 % and 11.40 % in C150 concrete. The same conclusion can be drawn from Fig. 2 and Fig. 3, the smaller the W/C ratio is, the denser the pore structure. The water in concrete is used as a basic medium for dissolving the cementitious materials during the hydration, and then participates in the reaction as a reactant. When the W/C is larger, the amount of cementitious material is less, which leads the cementitious material cannot completely wrap the surface of the aggregate. At this time, free water wraps the surface of fine aggregate, causing many pores to form inside concrete under unbalanced the action of pore water pressure. It will lead to the adjacent small pores connected into larger holes, forming a water-bleeding channel. Ultimately, the porous structure of concrete is formed. To a certain extent, it affects the pore diameter distribution of concrete. Moreover, with the increase of curing ages, the total pore volume and the volume of much harmful pores gradually decrease, while the volume of harmless pores gradually increases. The proportion of much harmful pores volume in total pore volume of C60 concrete from 27.61% at 7 days to 17.79% at 28 days, and 15.15% at 90 days, while the harmless pores from 20.20% at 7d to 28.15% at 28 days, and 39.06% at 90 days. The hydration process of C10-C150 concrete gradually stabilizes after 28

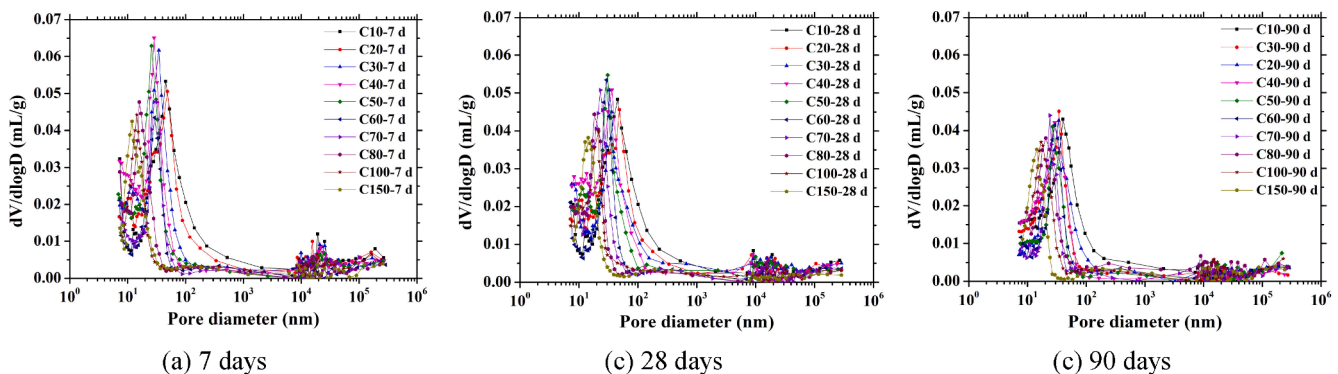


Fig. 2. Incremental pore diameter distribution curves of concrete at 7 days, 28 days and 90 days.

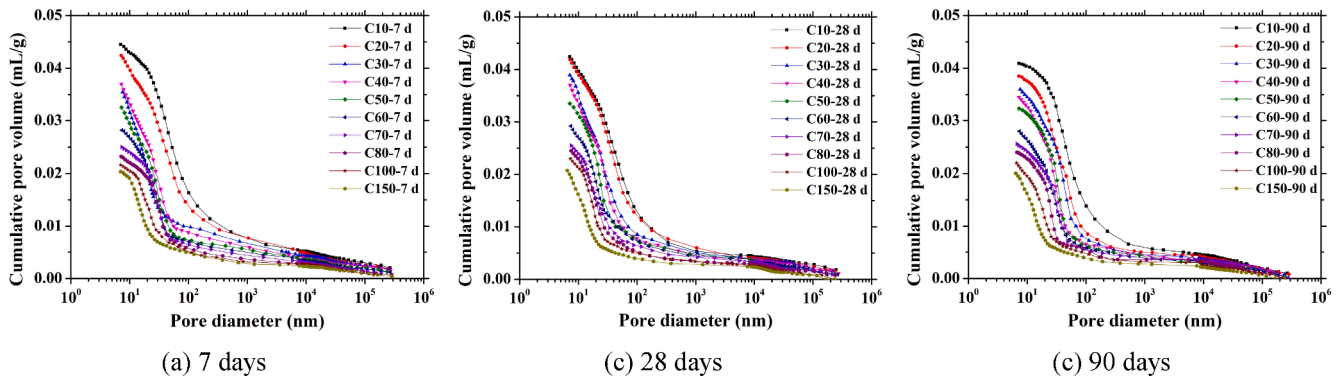


Fig. 3. Cumulative pore diameter distribution curves at 7 days, 28 days and 90 days.

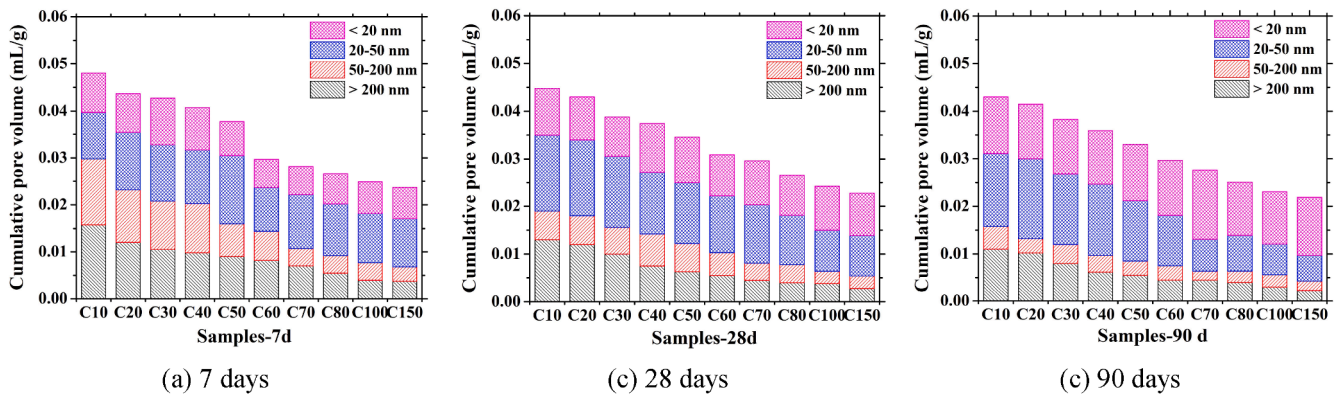


Fig. 4. Pore volume distribution of concrete at 7 days, 28 days and 90 days.

days. The main changes were that the volume of little harmful pore decreases and harmless pore increases at 90 days, which are similar to Fig. 2 and Fig. 3.

4.3. Specific surface area distribution

Fig. 5 Display the specific surface area of C10-C150 concrete at 7 days, 28 days, and 90 days. The harmless pores dominate the total specific surface area in all cases, followed by the little harmful pores. It reveals that the specific surface area depends largely on the volume of the micropores, and it has no obvious relationship with the volume of the mesopores and macropores. In addition, with the decrease of W/C, the total specific surface area and the specific surface area of little harmful pores shows an increasing trend. The proportion of harmless

pore volume in the total specific surface area of C30, C60 and C150 concrete is 36.46%, 39.47% and 41.32% at 7 days, 40.36%, 42.74% and 43.41% at 28 days, and 45.01%, 47.11% and 53.61% at 90 days. From Fig. 2- Fig. 4, the pore volume of the harmless pores increases with the decrease of W/C, while the pore diameter decreases. It leads to an increase in the roughness of pore surface area. However, the much harmful pores and harmful pores have no obvious change trend. Although, the pore volume of the harmless pores increases with the decrease of W/C, the pore diameter distribution range concentrates in the area with smaller values. The above phenomenon shows that the pore volume and pore diameter decide jointly the value of the pore specific surface area. When the pores volume is constant, the pore specific surface area increases with of pore diameter reduces. Therefore, the specific surface area of the pores can only be used as a reference value to describe the

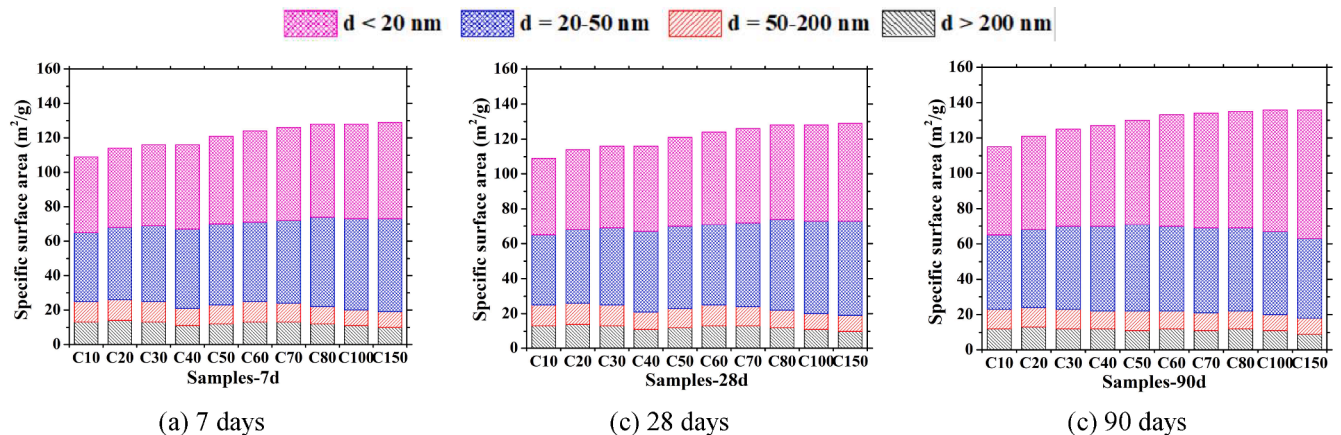


Fig. 5. Specific surface area distribution of concrete at 7 days, 28 days, and 90 days.

compactness of the pore structure and cannot be used as measure standards.

4.4. Fractal dimension and fractal regions

Due to the excessive number of fractal graphics in all experimental groups, Fig. 6 only shows the fractal curves of C10, C30, C60, C80, C100 and C150 concrete at 7 days, 28 days and 90 days. It could clearly

illustrate the scale dependence of fractal characteristics. Triplicate specimens were tested for each concrete mixture, and their average value were taken as the final value of the fractal dimension. The values of all fractal dimension are listed in Table 6 and Fig. 7. Fig. 8 illustrates the pore diameter ranges corresponding to all fractal regions. The fractal curves of C10-C150 concrete are composed of two parts. The first part corresponds to the low-pressure intrusion stage ($P = 0.0016\text{--}0.14\text{ MPa}$). In this stage, the mercury intrudes into cracks and macroscopic pores

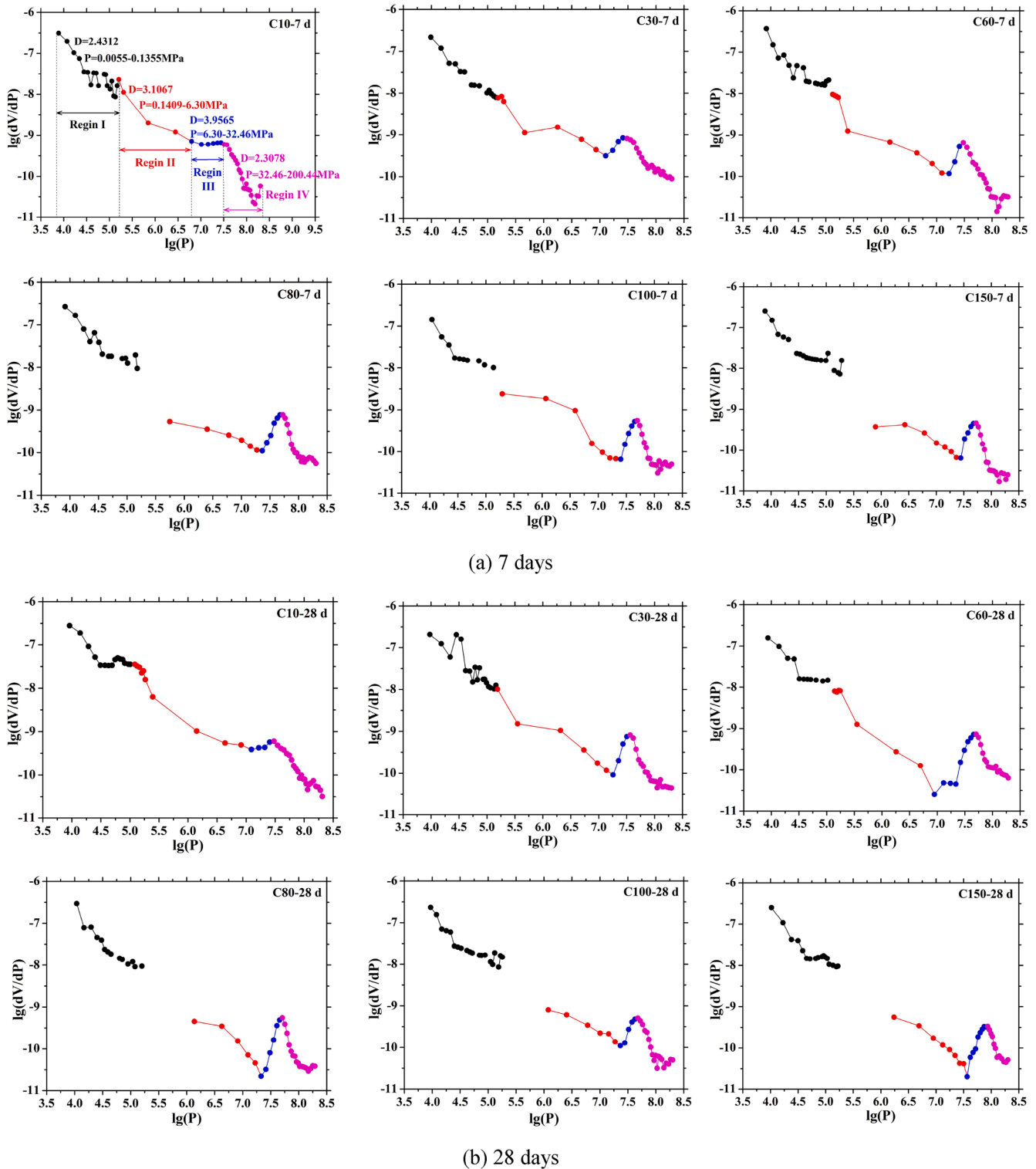


Fig. 6. The fractal regions of C10, C30, C60, C100, and C150 concrete.

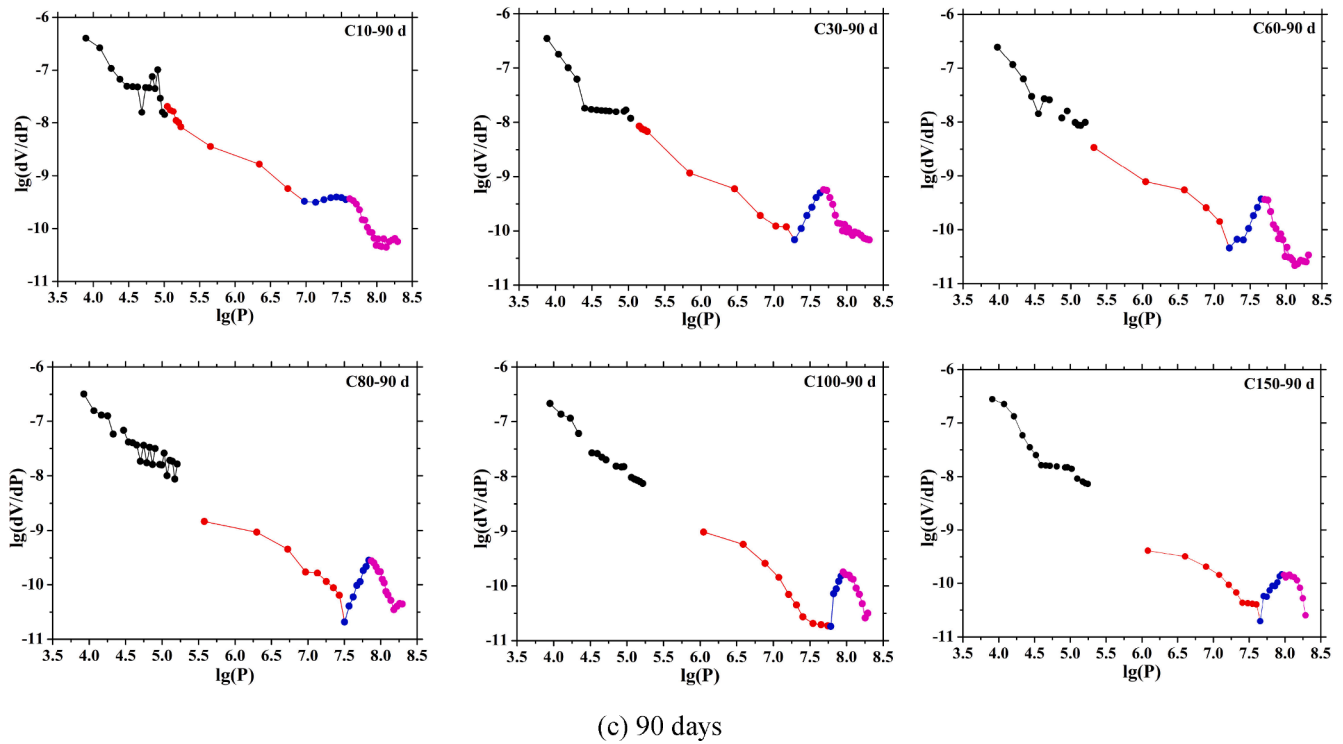


Fig. 6. (continued).

Table 6
The fractal Dimension of C10-C150 concrete.

Samples	7 d				28 d				90d			
	I	II	III	IV	I	II	III	IV	I	II	III	IV
C10	2.4312	3.1064	3.9565	2.3078	2.4653	3.3019	3.9061	2.3659	2.4867	3.2306	4.7694	2.4013
C20	2.4467	3.1494	4.5172	2.3337	2.4794	3.1556	4.5274	2.4101	2.5083	3.0711	6.4174	2.4274
C30	2.4673	3.3453	4.1105	2.3657	2.4992	3.3054	5.1550	2.4362	2.5194	2.8413	7.0265	2.4522
C40	2.4838	3.1244	4.9727	2.3786	2.5054	3.2443	6.0771	2.4555	2.5292	2.9182	7.9029	2.4654
C50	2.4998	3.1202	5.7721	2.4176	2.5105	3.1220	7.7216	2.4751	2.5354	2.9457	6.0515	2.5041
C60	2.5012	2.9246	7.2879	2.4692	2.5225	2.9361	4.2091	2.4954	2.545	2.9678	7.1494	2.5215
C70	2.5249	2.8522	7.2148	2.4824	2.538	2.5823	5.2543	2.517	2.5725	3.14	7.1276	2.5464
C80	2.5431	2.4647	6.6099	2.4946	2.5596	2.4574	6.3059	2.5389	2.5778	2.8319	6.2155	2.5562
C100	2.54553	2.4359	6.4261	2.5233	2.5632	2.4553	4.4015	2.5601	2.5896	3.2037	6.2178	2.5722
C150	2.5694	3.5505	6.9543	2.5334	2.5857	3.5905	4.9044	2.5771	2.6032	3.2306	4.7694	2.5845

which are surface opened. The second part corresponds to high mercury intrusion pressure ($P = 0.14\text{--}228\text{ MPa}$), the mercury starts penetrating throughout the sample into microscopic pore. There are two obvious peaks on the second part of each fractal curve. The diameter corresponding lowest peak on the fractal curve is threshold pore diameter, while the highest point is the most probable pore diameter.

From Fig. 6 and Table 6, obvious fractal features are revealed in the internal pore structure of concrete. However, owing to irregularity of pore distribution and diversity of pore morphology in concrete, it has the strongly scale-dependent and multi-fractal characteristics. It can be clearly observed that the second part of the line segment has different change trend in different pressure ranges, in Fig. 6. Thus, the fractal curves of each group concrete cannot be fitted with only one linear regression, but it needs a multi-linear regression. To clearly describe this characteristic, all Fractal curves based MIP and Menger sponge is usually divided three part according to the linear segmentation (linear fitting coefficient), namely microscopic (Region IV, within a fewdozen nanometers), macroscopic (Region I, over a few microns) and non-fractal regions (Region II and Region III). This also reflects the fact that hydrated particles exhibit different accumulation forms at different scales. Region I indicate the distribution of voids caused by drying shrinkage

and temperature changes in the system, bubbles caused by uncompacted vibration during forming, and larger pores formed during hydration. Region II is the region of the pores between the diameter corresponding to the initial pressure value of high-pressure and the threshold pore diameter. Region III represents the fractal region from the threshold diameter to the most probable diameter. The amount of mercury injected increases obviously in this region. Region IV mainly represents the fractal curves of pores that below the most probable diameter. Region I is macropores fractal region ($d = 5\text{ }\mu\text{m} \sim 750\text{ }\mu\text{m}$), represents the packing patterns of hydrated cement grains, while Region II - Region IV is micropores fractal region ($d = 5\text{ nm} \sim 5\text{ }\mu\text{m}$), reflects mainly the microstructure of C-S-H hydrates.

From Fig. 6 and Fig. 8, the scale of fractal region of each group concrete is different. Region I and Region IV exhibit obvious fractal features, the fractal dimension D is between 2 and 3. The changing trend of fractal dimension of Region I is the same as Region IV. When the curing age lengthens, the pore pressure ranges and pore diameter ranges reduces gradually, the fractal dimension increases. With the progress of the hydration reaction, hydration products are gradually formed, and the packing pattern of hydrated cement grains is denser in the concrete. The pore structure is more refined. It will lead to pore surface area

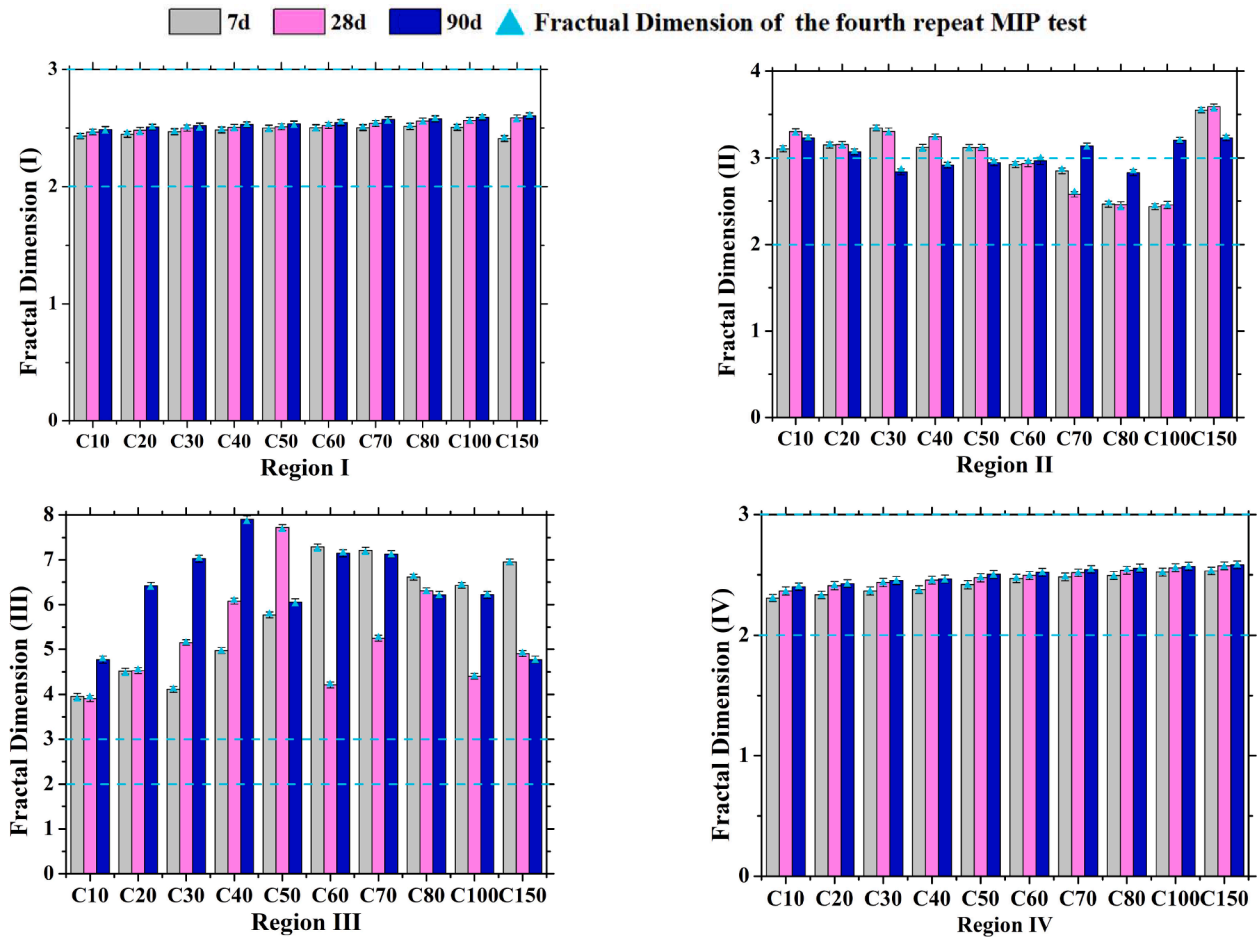


Fig. 7. The fractal Dimension and Standard Deviation of C10-C150 concrete.

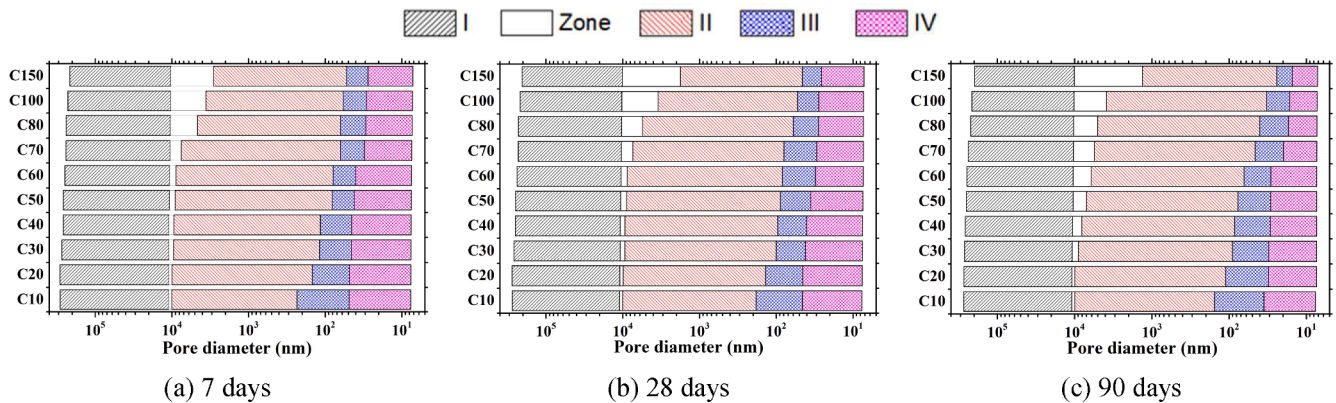


Fig. 8. The pore diameter in Region I and Region IV at 7 days, 28 days and 90 days.

increases and rougher internal surface and more complex pore structure of concrete. Therefore, the fractal dimension gradually increases and stabilizes in the later stage of the hydration reaction [28]. Else, the pore diameter decreases with the decrease of W/C, and the fractal dimension shows an opposite trend. When the W/C is small, it is equivalent to an increase in the amount of cement in the system. It can make the surface of aggregate in concrete completely wrapped. Except that, cement can play the role of physical filling, and it well can fill the coarse pores at the aggregate interface. Both of these effects can refine the internal pore structure of concrete. This conclusion can also be drawn from Fig. 2- Fig. 5, and other researchers have proposed similar conclusions

[11,26,33,35-37]. However, the fractal dimension of Region I is always greater than Region IV, they are between 2.4312 and 2.6032 and 2.3078-2.5845, respectively. This reveals extremely irregular and disordered packing patterns of hydrated cement than the pore structure of C-S-H gels [26,31,33,37].

However, the fractal dimension $D > 3$ in Region III, which are non-physical from the point of view of surface geometry. The pore ranges of Region II and Region III reduce with the increments of the curing age, and the pores are more refined, drifting from the micrometer scale to the nanometer scale. But with increases of W/C, it showed the opposite trend. It is worth noting that the fractal region and non-fractal regions of

Region II are unfixed, evolves in different ranges with curing time. Region II is the region of the pores between the diameter corresponding to the initial pressure value of high-pressure and the threshold pore diameter, including the large capillary pores of 100 nm–5 μm . It is known that the strength of hardened cement composite cementitious materials shows a high correlation with the amount of large capillary voids exceeding 100 nm. But the pore ranges in Region II varies between micrometer scale to nanometer scale with curing age, including large capillary pores and fine capillary pores < 100 nm. In addition, the fractal dimension is not only related to pore diameter and porosity, but also closely related to pore morphology. Thus, the scale of fractal region and non-fractal region of Region II is not fixed, but evolves in different ranges with the passage of time. This also shows that hydrated particles show different stacking forms at different scales. Else, noted that the important jumps between Region I and Region II, which is a transition region with no fractal property. It may be the result of the assumption of Washburn formula in the MIP is too idealizes. Firstly, the MIP assumes that the pores of the samples to be measured are all open pores. However, there are close pores and half-open pores in the actual samples. It results in some pores that cannot be measured. Furthermore, the relationship between pressure and volume of mercury injected in the MIP is based on the cylindrical pore model, but there are many kinds of pore morphology in the actual samples. It is presumed that the generation of fractal regions and non-fractal regions of pores ranges is not only related to the pore structure, but there may be a potential correlation with pore morphology, such as ink bottles pores [33]. When the samples contain a large number of pores morphology like ink bottles, it is easy to cause part of the mercury to remain in the samples, forming a phenomenon of mercury hysteresis. Fig. 9 shows that the compared the pore size distribution and pore volume between the pressurization and decompressive process. It can be seen from Fig. 9, the curves of decompressive did not cover that of pressurization, and there was a phenomenon of retention and hysteresis during decompressive. The above phenomenon of non-fractal and its reason has also been shown in some literature [26,31,33,34].

Overall, fractal curves and fractal dimensions can well describe the spatial distribution and internal features of pores in concrete. However, for the fractal curves measured by MIP, the self-similarity and scale-dependency of pore surface for concrete exist only within certain ranges. Therefore, it is necessary to choose a reasonable range of pore diameter when analyzing the internal pore structure of concrete by MIP.

4.5. The link between fractal dimension and pore structure characterization parameters

The characterization parameters of pore structure in concrete include pore volume, pore diameter distribution, specific surface area,

and pore morphology, etc.. However, it is not accurate to express the pore structure only by a single parameter. For instance, the pore volume is an overall summary value to indicate the total number of pores inside the material, which cannot accurately describe the number and diameter of pores in the concrete. The pore diameter in concrete ranges from Å to μm , and different pore diameters have different effects on various aspects of the material's properties. The most probable pore diameter is used to represent the cracking width of the largest number of pores in the concrete. It can be used to analyze the change of pore diameter. The pores specific surface area can only characterize the surface roughness of pores surface to a certain extent. Moreover, the most probable pore diameter and pore specific surface area are calculated under the premise of assumption, which only has theoretical significance. Many literatures [19,38] draws the conclusion that the pore structure of concrete can be considered as a fractal object in the aspect of pore surface areas, pore volumes, pore diameter, etc.. Therefore, it is very necessary to analyze the pore structure parameters of concrete by fractal theory.

In Fig. 6, the concrete has fractal feature only in Region I and Region IV. In this paper, the relationship between the fractal dimension and the most probable pore diameter, porosity, and pore specific surface area in Region I and Region IV only is considered, and displayed in Fig. 10 and Fig. 11. From Fig. 10, the relationship between the pore volume and the fractal dimension is discretely distributed in Region I, while the distribution shows a negative correlation in the Region IV. However, the correlation coefficient between pore volume and fractal dimension is 0.7341, which is smaller than that of fractal dimension and the most probable pore diameter and pore specific surface area. This illustrates that the complexity of pore structure distribution is also affected by other factors, such as pore diameter and specific surface area, when the pore volume is the same. The relationship between the fractal dimension and the most probable pore diameter both shows a negative correlation, while the relationship with the specific surface area shows a positive correlation, in Region I and Region IV. This is cause that when the pores inside the concrete are mainly medium and small pores, the surface roughness of the pores increases, and thus the fractal dimension increases. This is consistent with the fractal theory. Moreover, the correlation in the Region I is less than that in the Region IV, which can be understood as the ability of micropores to occupy space is relatively strong.

Fig. 11 shows the relationship between fractal dimension and pore structure characterization parameters based on different curing days. It is clear that the development trend of the relationship between fractal dimension and pore structure characterization parameters in C10-C150 concrete is the same at 7 days, 28 days, and 90 days, as well the development trend in each age is the same as the overall trend. In Region I and Region IV, the relationship between the fractal dimension and the specific surface area both show a positive correlation, while the

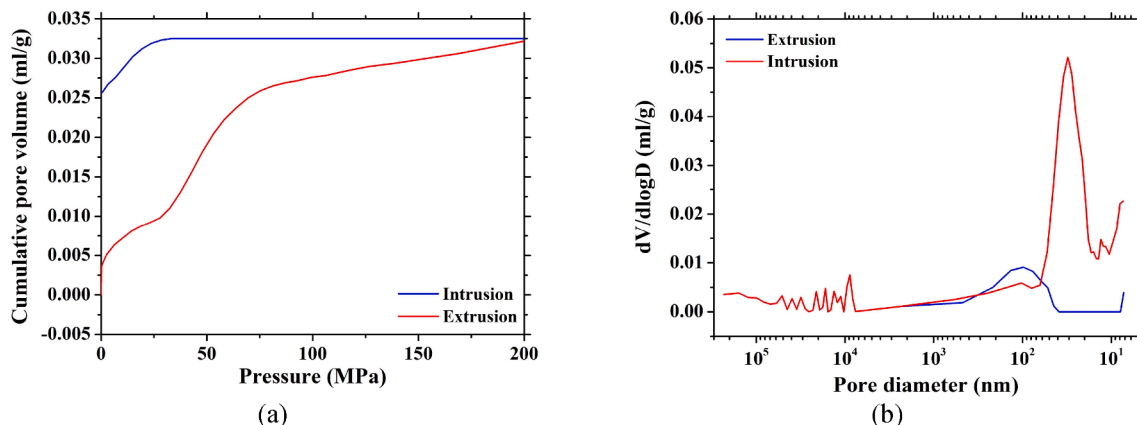


Fig. 9. The curves of mercury-intrusion and mercury-extrusion.

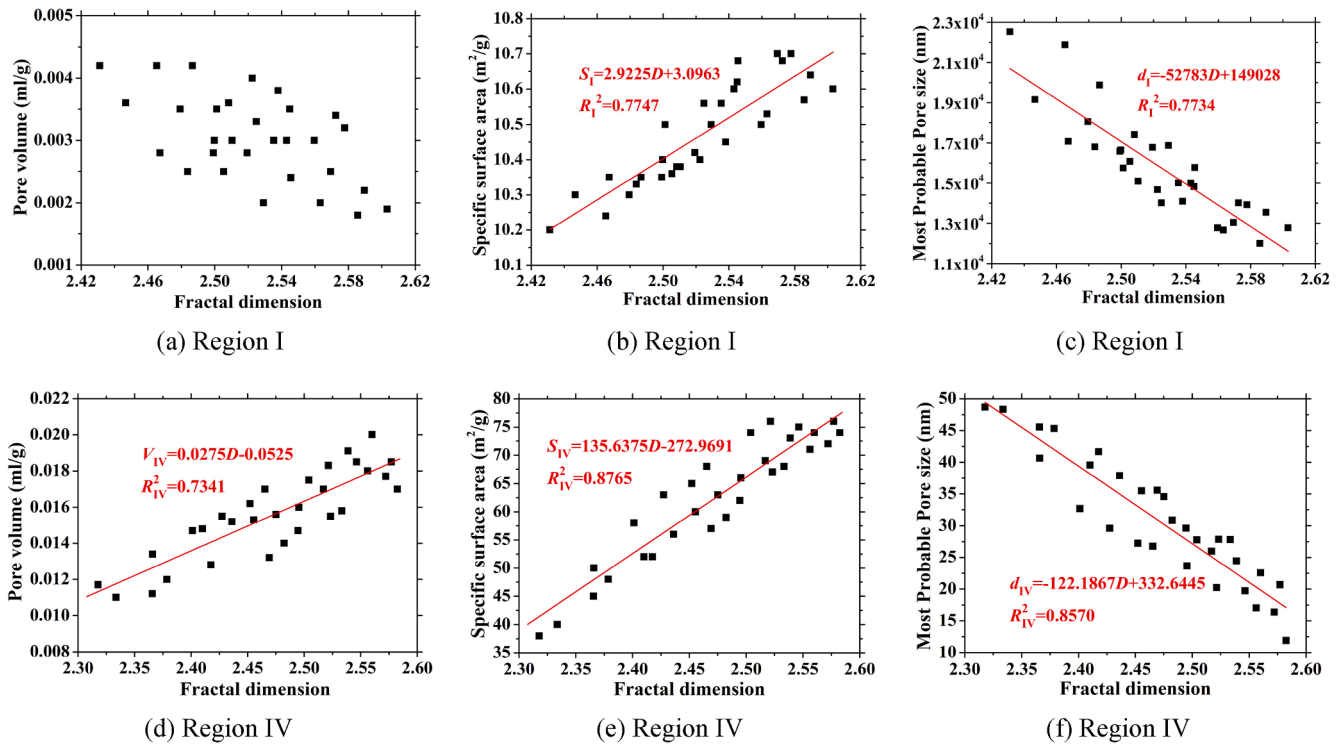


Fig. 10. The link between fractal dimension and pore structure characterization parameters.

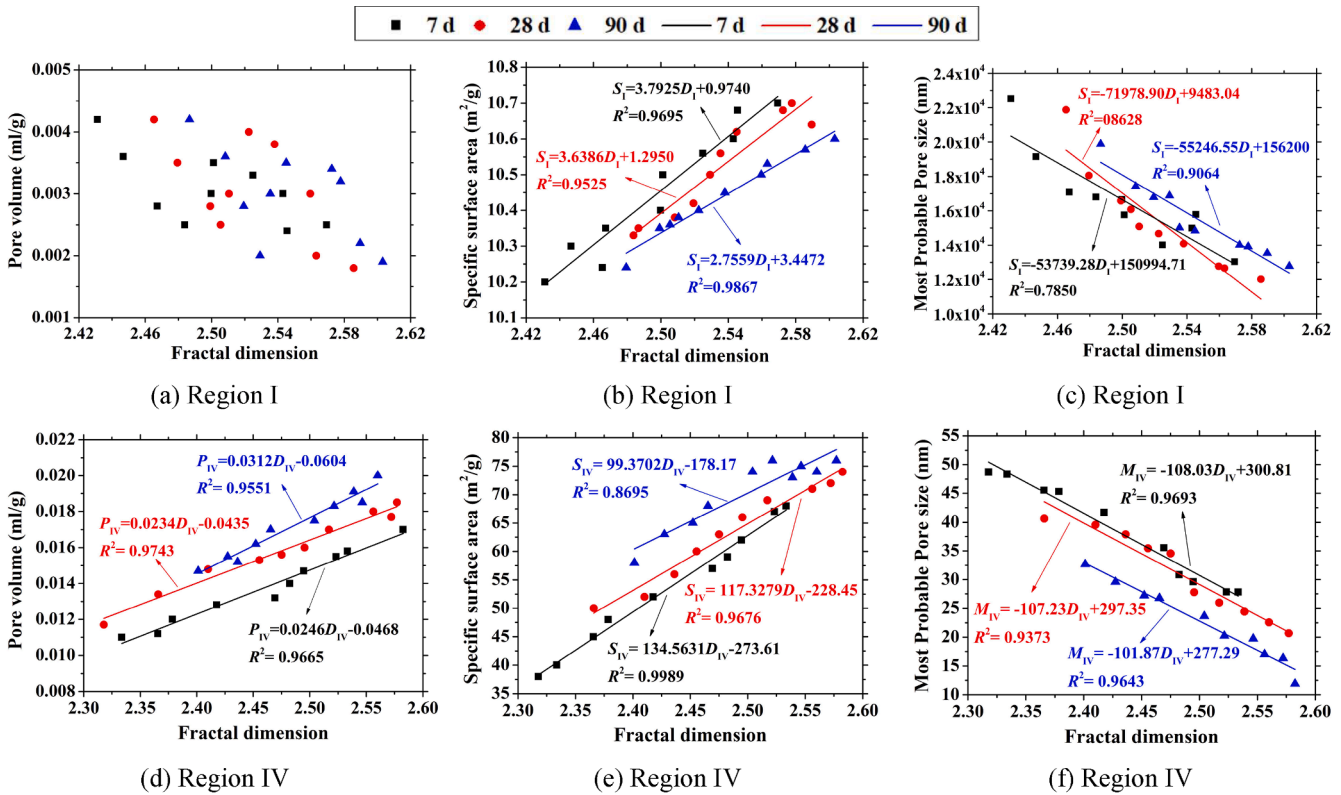


Fig. 11. The link between fractal dimension and pore structure characterization parameters.

relationship with the most probable pore diameter shows a positive correlation at 7 days, 28 days, and 90 days. The relationship between the pore volume and the fractal dimension is discretely distributed in Region I, while the distribution shows a negative correlation in Region IV. This

proves once again the conclusion that the pore structure of concrete can be considered as a fractal object in the aspect of pore surface areas, pore volumes, pore diameter and the ability of micropores to occupy space is relatively strong.

According to the above analysis, the fractal dimension can clearly describe the internal pore characteristics of pores. Meanwhile, it has a good correlation with the pore volume, the most probable pore diameter, and the pore specific surface area. The pore volume, the most pore diameter and pore specific surface area of concrete can be inferred by comparing the dimensions of the fractal dimension. Compared with other pore structure characterization parameters, pore volume fractal dimension can be considered as a comprehensive parameter of pore morphology and pore spatial distribution, which can accurately reflect the pore structure characteristics.

4.6. Compressive strength

Fig. 12 displays the observed values of compressive strength of C10-C150 concrete, to better meet the design requirements. In Fig. 12, $f_{cu,0}$ (min) and $f_{cu,0}(\min)$ are the minimum values of compounding strengths of C10-C50 concrete and C60-C150 at 28 days, respectively. And $f_{cu,k}$ is the standard value of concrete cube compressive strength at 28 days. It shows a well-known law obviously that the W/C is an important factor in determining strength. Concrete is heterogeneous composite materials composed of crystals, gels, aggregates, incompletely hydrated particles, free water, and pores. The action of crystal, gel and incompletely hydration particles is to increase the intensity, while free water and pores is the opposite. When the W/C is relatively small, free water in the system decreases, crystals and gels increase, thereby increasing the strength of the concrete. When water consumption exceeds the requirements for

cement hydration, excessive free water will migrate and bleed, which will produce a large number of pores in concrete. It will lead to the force-bearing area of the concrete specimen diminish. Meanwhile, the strength of the interface between the aggregate and the cementitious materials reduces. Hence, the compressive strength of concrete is closely related to the density of pores structure.

4.7. Mathematical model

When analyzing the relationship between pore structure and strength, many parameters need to be considered. But it will lead to the mathematical model expression too complex. Hence, the representative parameters were selected for a simple model from upper parameters. Combined with analysis in 4.4 and 4.5, pore volume fractal dimension can be considered as a comprehensive parameter of pore morphology and pore spatial distribution, which can accurately reflect the pore structure characteristics. Import the data of compressive strength and fractal dimension measured by MIP test into Origin. The scatter plot was drawn for the data of fractal dimension and compressive strength by Origin. Finally, the mathematical model expression of compressive strength and fractal dimension was obtained by linear regression analysis of scatter diagram. The mathematical model expression between pore volume fractal dimension and compressive strength be established in Fig. 13. The F-test was used to reveal the significance of the regression constants and function [38]. The value of regression and significance were listed in Table 7.

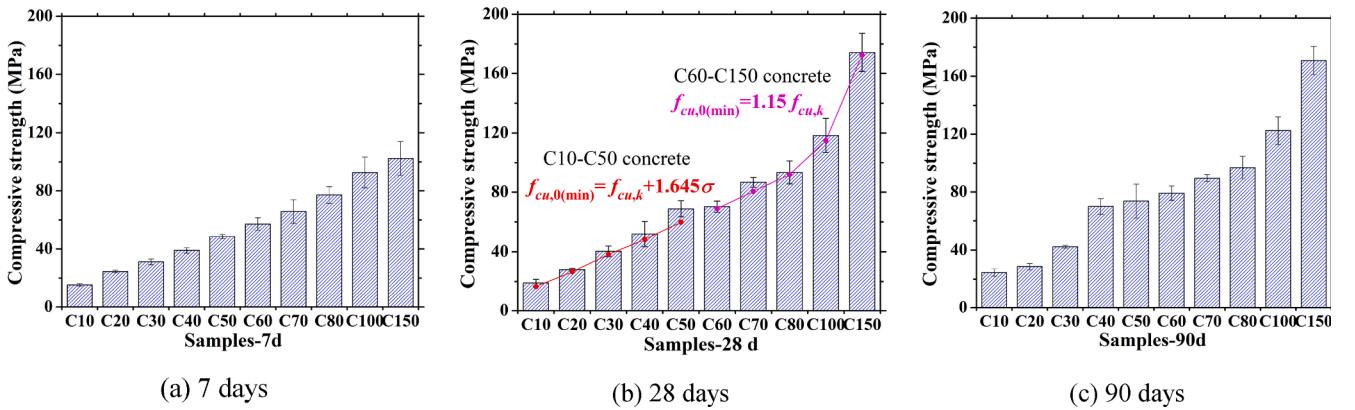


Fig. 12. Compressive strength and standard deviation of concrete.

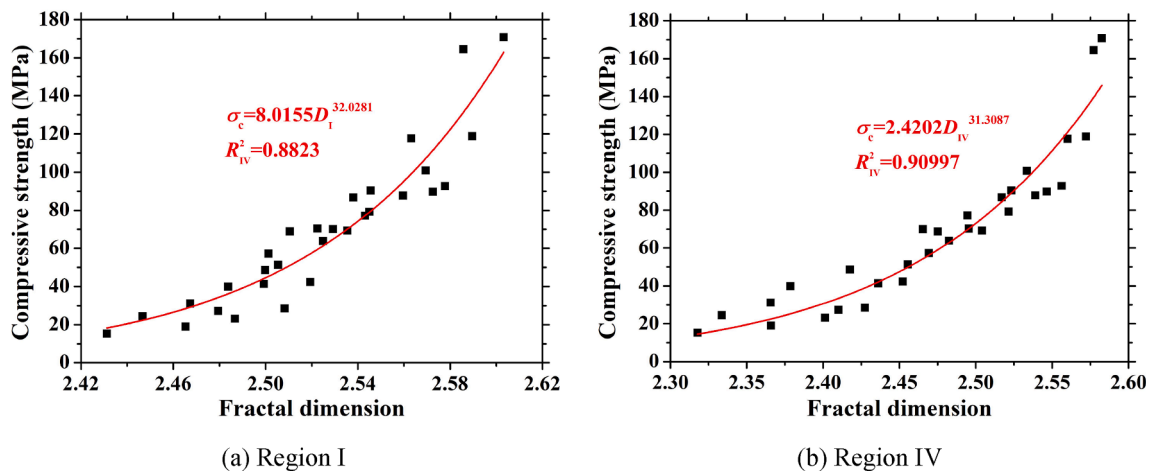


Fig. 13. The mathematical model between fractal dimension and compressive strength.

Table 7

The values of Regression Regression and significance.

Regression equations	<i>a</i> Value	Standard Error	<i>b</i> Value	Standard Error	Statistics Reduced Chi-Sqr	<i>R</i> ²	F Value
$\sigma_c = aD_I^b$	8.0155	1.8214×10^{-11}	32.0281	2.4154	183.2224	0.8823	498.3083
$\sigma_c = aD_{IV}^b$	2.4202	3.3933×10^{-7}	21.3087	1.5070	140.1461	0.9100	655.7751

In Fig. 13, the fractal dimension of Region I and Region IV both have a good correlation with the compressive strength, according to a positive power function relationship which is expressed as follows Eq. (9) and Eq. (10).

$$\sigma_c = aD_I^b \quad (9)$$

$$\sigma_c = aD_{IV}^b \quad (10)$$

Where σ_c is Compressive strength of concrete; D_I and D_{IV} is the fractal dimension of Region I and Region IV; a and b are regression constants. R^2 I and R^2 IV is the correlation coefficient Region I and Region IV.

The fractal dimension of Region I and Region IV both have a good correlation with the compressive strength, the larger the fractal dimension is, the greater the compressive strength of concrete is. It can be seen from Fig. 6, Table 6 and Fig. 7, The changing trend of fractal dimension of Region I is the same as Region IV. When the curing age lengthens, the pore pressure ranges and pore diameter ranges reduce gradually, the fractal dimension and the compressive strength increases. In other words, with the progress of the hydration reaction, hydration products are gradually formed, and the packing pattern of hydrated cement grains is denser in the concrete, and the compressive strength is higher. At this time, the pores in concrete are mostly concentrated in the range of small pores. It will lead to pore surface area increases and rougher internal surface and more complex pore structure of concrete. Therefore, the fractal dimension gradually increases. This means that the larger the fractal dimension is, the more complex the spatial geometric characteristic of the pore structure is. Thus, the stronger the space filling capacity of concrete, the higher the compressive strength. This proves once again that the magnitude of compressive strength is closely related to the compactness of microscopic pore structure. However, the correlation coefficient of R^2 IV is greater than that of R^2 I, which are 0.9112 and 0.8823. This is because formation methods of different pores are different, the multifractal characteristics of concrete and the correlation coefficient of fractal dimension are also different. In addition, the fractal dimension of Region I is always greater than Region IV. This reveals extremely irregular and disordered packing patterns of hydrated cement than the pore structure of C-S-H gels. Therefore, the relationship between fractal dimension and compressive strength of different pores is also different.

Overall, this mathematical model in Region I and Region IV both expresses the relationship between macroscopic mechanical properties and microscopic pore structure indexes and the way of mutual influence between them. However, comparing Region I and Region IV, the correlation between fractal dimension and compressive strength is better in the latter. Therefore, it is more precise to choose the mathematical model in Region IV to predict the strength of denser concrete.

5. Conclusions

- (1) The internal pores density and compressive strength of concrete increase with decrease of W/C or increase of curing age. The hydration process of concrete gradually tends to be stable, and the most probable pore diameter and pore volume reduced after curing 28 days, then the pores are mostly concentrated in the less harmful pores and the harmless pores, results in the slow growth of compressive strength of concrete.
- (2) Obvious fractal features are revealed in the internal pore structure of concrete. All fractal curves are divided into four segments

according multi-fractal characteristics of concrete, namely Region I, Region II, Region III, and Region IV. Region I is macropores fractal region ($d = 10 \mu\text{m} \sim 950 \mu\text{m}$), represents the packing patterns of hydrated cement grains, while Region II - Region IV is micropores fractal region ($d = 0.0064 \mu\text{m} \sim 10 \mu\text{m}$), reflects mainly the microstructure of C-S-H hydrates. Region I and Region IV exhibit obvious fractal features, the fractal dimension D is between 2 and 3. However, the fractal dimension $D > 3$ in Region II and Region III, which are non-physical from the point of view of surface geometry.

- (3) The mercury intrusion porosimetry is a very effective and practical method for measuring the pore diameter distribution of concrete. However, due to the deviation of the morphology of pores from the ideal pore model and the limitation of pressure range in the actual samples, not all the pores conform to the fractal features.
- (4) The internal pore characteristics of pores can be clearly described by fractal dimension. And fractal dimension has a good correlation with the most probable pore diameter and the pore specific surface area. The pore volume fractal dimension can be considered as a comprehensive parameter of pore morphology and pore spatial distribution, compared with other pore structure characterization parameters, which can accurately reflect the pore structure characteristics.
- (5) There is a good correlation between the fractal dimension of Region I and Region IV and the compressive strength, the mathematical model according with a positive power function relationship. Compressive strength can be inferred by comparing and analyzing the fractal dimension in the concrete. In practical engineering, the damage detection of concrete on site may be realized by the mathematical model.

6. Further research

- (1) More cement composite cementitious materials with different mix proportions should be considered in further, to verify and modify the mathematical model between the compressive strength and the fractal dimension.
- (2) In this test, the pore structure of concrete was measured by the mercury intrusion porosimetry(MIP). Further, the pore structure with various cement composite cementitious materials will be measured by different methods, such as nitrogen adsorption method, small angle x-ray scattering (SAXS), scanning electron microscope (SEM), Computed Tomography (CT) and different calculation models, the similarities and differences between them will be attempted to compare and analyze.
- (3) It is a simplified mathematical model between the fractal dimension and compressive strength in this experiment, and it is in the stage of exploration and research at present. In the future, the relationship between the fractal dimension and compressive strength will be further analyzed to establish a more accurate mathematical calculation model.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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